

Role of maca (*Lepidium meyenii*) consumption on serum interleukin-6 levels and health status in populations living in the Peruvian central Andes over 4000 m of altitude

Lepidium meyenii (Maca) is a plant that grows at over 4000 meters above sea level in the central Peruvian Andes. The hypocotyls of this plant are traditionally consumed for their nutritional and medicinal properties. The aim of this study was to determine the health status based on a health related quality of life (HRQL) questionnaire (SF-20) and serum levels of interleukin 6 (IL-6) in subjects that are maca consumers. For this, a cross-sectional study was designed to be performed in 50 subjects from Junin (4100 m): 27 subjects were maca consumers and 23 were non-consumers. The SF-20 survey is used to obtain a summary measure of health status. The stand up from a chair and sit down (SUCSD) test (to assess lower-extremity function), hemoglobin measurement, blood pressure, sexual hormone levels, serum IL-6 levels and the score of chronic mountain sickness (CMS) were evaluated. Testosterone/estradiol ratio ($P=0.05$), IL-6 ($P<0.05$) and CMS score were lower, whereas the health status score was higher, in maca consumers when compared to non-consumers ($P<0.01$). A greater proportion of maca consumers successfully completed the SUCSD test compared to non-consumers ($P<0.01$), showing a significant association with lower values of serum IL-6 ($P<0.05$). In conclusion, consumption of maca was associated with low serum IL-6 levels and in turn with better health status scores in the SF-20 survey and low chronic mountain sickness scores.